

I-CHUN CHEN

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🐙 <https://github.com/wazenmai>

🌐 <https://blog.wazenmai.com>

My research interests broadly lie in the fields of image/text/music AI, model compression, and explainable AI. I have a paper accepted at ICML 2025 related to Large Language Models (LLMs), specifically focusing on Mixture-of-Experts (MoE) architecture and compression. Recently I'm working on VLM inference and applications.

EDUCATION

National Tsing Hua University

September 2022 - July 2024

Master's degree. Department of Computer Science.

- Overall GPA: 4.23/4.3

Technische Universität Dresden, Germany

March 2022 - August 2022

Exchange Student of Computer Science.

National Tsing Hua University

September 2018 - August 2022

Bachelor's degree. Department of Computer Science.

- Overall GPA: 3.93/4.3

PUBLICATIONS

Retraining-Free Merging of Sparse Mixture-of-Experts via Hierarchical Clustering

🌐 <https://arxiv.org/abs/2410.08589>

🐙 <https://github.com/wazenmai/HC-SMoE>

- Accepted by Forty-Second International Conference on Machine Learning (ICML 2025) as first author.
- Reducing parameter of experts in MoE in retraining-free, task-agnostic manner.
- Utilize hierarchical clustering to group similar experts with a merging process within ten minutes.
- Evaluate Mixtral 8x7B and Qwen1.5-MoE on eight language benchmarks.

MidiBERT-Piano: Large-scale Pre-training for Symbolic Music Understanding

🌐 <https://arxiv.org/abs/2107.05223>

🐙 <https://github.com/wazenmai/MIDI-BERT>

- Accepted by Journal of Creative Music Systems (JCMS), 2024.
- Apply BERT into general musical tasks such as melody completion, composer classification and emotion classification using Pytorch.
- Research for two kinds of music representation, REMI and CP and study the performance difference between those results.

WORK EXPERIENCE

Cyberlink

December 2024 - Now

RD-AI Engineer

- Trained a human parsing model based on Segformer that achieves mIoU 77.03 on the validation set of DeepFashion.
- Implemented OpenVINO and ONNXRuntime C++ inference pipeline for human parsing.
- Surveyed SOTA VLMs (DeepSeek, QwenVL-2, Ola, Ovis-2) and improve VLM inference throughput around 1.6 times through vLLM.
- Implemented auto quality evaluator by VLM(Gemma-3) in Python.
- Constructed video editing pipeline with Python, ffmpeg, VLM(Gemini) and Text-to-Speech model(IndexTTS).

Cloud Communication Technology LTD.

February 2022 - August 2023

iOS and Android App Engineer

- Implemented LINE message API and login service in Community Web Flutter app.
- Implemented version detection, encrypted qrcode scanner, Google Cloud Storage Image upload and deeplink function in TABF Exam Camera Flutter app.
- Embed Google face detector in camera so we can detect the face angle and position in real-time for ID Photo camera Flutter app.

- Designed and Implement backend API and database CRUD operations for ID Photo camera with typescript , Koa and PostgreSQL.

Research Center for Information Technology Innovation, Academia Sinica

July 2020 - August 2020

Music & AI Lab Summer Intern

- Learned symbolic music format representation and encoding.
- Presented papers in NLP transformer family, including Transformer, BERT, RoBERTa, ELECTRA, GPT-2, and T5.
- Modified "Pop Music Transformer" from single-track piano music generation to multi-track pop music generation, the model can either generate songs by itself or continuation after human-given music.

PROJECTS

2022 AI-CUP Explanation Information Tagging of Natural Language Understanding Competition

🔗 <https://github.com/wazenmai/DataMining-Team5>

- Given the talk between two people and one of them's attitude (agree or disagree), we need to find the important statement among the sentences they said to support their main point and attitude.
- View problem as token classification task and fine-tune on BERT, DistillBERT and RoBERTa, also as question answering task and fine-tune on BERT.
- Got score 0.835978 (1 is best and 0 is worst) from token classification RoBERTa with **30th place** in private leader board.

NTHU OAuth Decaptcha

🔗 <https://chrome.google.com/webstore/detail/nthu-oauth-decapcha/mflpajkffpiibelpmffonolenndbgogp>

🔗 <https://github.com/wazenmai/NTHU-OAuth-Decaptcha>

- Trained a CNN model to recognize the number on the captcha image, get 98.9% accuracy on testing set.
- Collected training data by ourselves by posting request with captcha id then we can easily get lots of captcha images.
- There are 356 users using our decaptcha extension.

Parallel Pitch Tracking Algorithm

🔗 <https://github.com/wazenmai/Parallel-Pitch-Tracking>

- Implemented ACF (Auto-Correlated Function) with MPI and CUDA to recognize pitch in the wav file.
- About 100x speedup with the largest testing data by CUDA optimization.

Massive Data Analysis

🔗 <https://github.com/wazenmai/Massive-Data-Analysis>

- Learned topics in massive data analysis, including MapReduce, similarity search, frequent-itemset mining, managing advertising and recommendation systems, etc.
- Using MapReduce in Python Spark on matrix multiplication, pagerank, kemans, and finding similarity articles.

TECHNICAL SKILLS

Programming:	C, C++, Python, Dart, Javascript, CUDA
Software & Tools:	ML-Related Framework: Pytorch, Huggingface-transformers
	Frontend Framework: Flutter, VueJS
	Cloud Service: GCP(Vision API, Cloud Storage), Firebase
	Database: PostgreSQL, MongoDB

EXTRACURRICULAR

Machine Learning Study Group

November 2019 - August 2021

- Watched and discussed ML-related videos every week, as well as implement algorithms and read papers.
- Including course of Machine Learning on Coursera by Andrew Ng, CS231n: Deep Learning of Computer Vision at Stanford University, Reinforcement Learning course by DeepMind David Silver, and CS294-112: Deep Reinforcement Learning at UC Berkely by Sergey Levine.

LeetCode Study Group - Convener

February 2021 - June 2021

- Formed a 16 students group and invite lecturer to explain 6 problems every 2 weeks.
- Practiced Leetcode problems by topics, including dynamic programming, binary search, stack, etc.
- Got the **Excellent Award** of NTHU 2021 winter semester study group.

🔗 <https://justin0u0.notion.site/LeetCode-57d3fc3c39714bf7b37429a590a85824>